

LAB 8

Server Configuration – DHCP, DNS and Web Server in Cisco Packet Tracer

OBJECTIVES

- To set up a server in Cisco Packet Tracer that provides DHCP, DNS, and HTTP (web) services.
- To understand how DHCP dynamically assigns IP configuration details to client devices.
- To learn the role of DNS in translating domain names into IP addresses within a network.
- To verify the proper functioning of these services by accessing the hosted web page from client PCs.

THEORY

1. Dynamic Host Configuration Protocol (DHCP)

DHCP is a network protocol that automatically provides IP addresses, subnet masks, gateways, and DNS details to client devices. This eliminates the need for manual configuration on each device and ensures efficient IP address management. In Cisco Packet Tracer, enabling DHCP on a server allows hosts to dynamically obtain their network settings, reducing administrative effort and preventing address conflicts.

2. Domain Name System (DNS)

DNS translates human-friendly domain names into numerical IP addresses that computers can understand. Without DNS, users would need to remember IP addresses to access network resources. When configured on a server in Packet Tracer, DNS enables clients to resolve domain names to the correct IP of the web server, replicating real-world internet functionality where DNS acts as a directory service.

3. Web Server (HTTP Service)

A web server stores and delivers web pages to clients using the HTTP protocol. When a user enters a domain name in a browser, DNS resolves it to an IP address, and the HTTP service provides the requested web content. In Packet Tracer, enabling HTTP on the server allows PCs to access and display web pages, confirming that DHCP and DNS services are operating correctly.

4. Integration of Services

DHCP, DNS, and HTTP services work together to ensure smooth network operation:

- DHCP assigns IP configuration to client devices automatically.
- DNS converts domain names into IP addresses.
- The web server delivers requested content to users.

NETWORK TOPOLOGY

The network follows a simple star topology in which all devices are connected to a central switch. The server acts as the core resource, providing DHCP, DNS, and web services, while the PCs function as clients to test connectivity and service availability.

This arrangement illustrates how centralized network services can efficiently manage IP addressing, domain name resolution, and web hosting within a small-scale network environment.

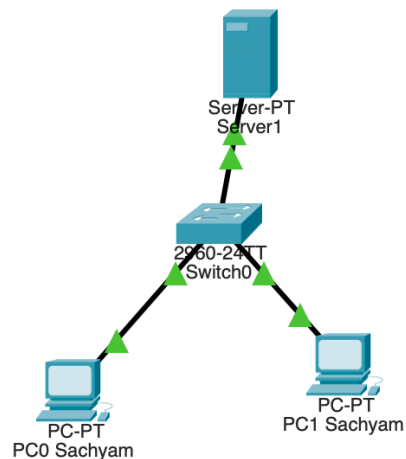


Fig: Network Topology

Configuration

IP ADDRESSING SCHEME

Device	IP Address	Subnet Mask	Gateway
Server	192.168.1.2	255.255.255.0	192.168.1.1
PC	DHCP	DHCP	DHCP

PC NETWORK CONFIGURATION (Obtained via DHCP)

Device	IPv4 Address	Subnet Mask	Default Gateway	DNS Server
PC0	192.168.1.1	255.255.255.0	0.0.0.0	192.168.1.2
PC1 Sachyam	192.168.1.3	255.255.255.0	0.0.0.0	192.168.1.2

SERVER CONFIGURATION

Step 1: Assign Static IP Address to the Server

1. Click on **Server** in Cisco Packet Tracer.
2. Navigate to **Desktop → IP Configuration**.
3. Configure the following parameters:
 - **IP Address:** 192.168.1.2
 - **Subnet Mask:** 255.255.255.0
 - **Default Gateway:** 192.168.1.1
 - **DNS Server:** 192.168.1.2

DHCP CONFIGURATION

Step 2: Enable DHCP Service

1. Go to **Server → Services → DHCP**.
2. Set **DHCP status** to **ON**.

Step 3: Create DHCP Pool

Field	Value
Pool Name	LAB_Pool
Default Gateway	192.168.1.1
DNS Server	192.168.1.2
Start IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Maximum Users	50

Click Add to save the configuration.

VERIFICATION OF DHCP

On each PC:

1. Go to **Desktop → IP Configuration**
2. Select **DHCP**

The PCs should automatically receive IP addresses from the DHCP pool.

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Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=2ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

```

Ping Test from PC1 Sachyam to PC0 Sachyam

DNS SERVER CONFIGURATION

Step 4: Enable DNS Service

1. Go to **Server** → **Services** → **DNS**
2. Turn **DNS ON**

Step 5: Add DNS Record

Name	Type	Address
www.hcoe.com	A Record	192.168.1.2

```

C:\>ping www.hcoe.com

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

Fig: Ping Verification on PC1 Sachyam

WEB SERVER CONFIGURATION

Step 6: Enable HTTP Service

1. Go to **Server** → **Services** → **HTTP**
2. Turn **HTTP ON**
3. (Optional) Edit **index.html** to customize the web page

VERIFICATION

On any PC:

1. Go to **Desktop** → **Web Browser**
2. Enter the URL: <http://www.hcoe.com>

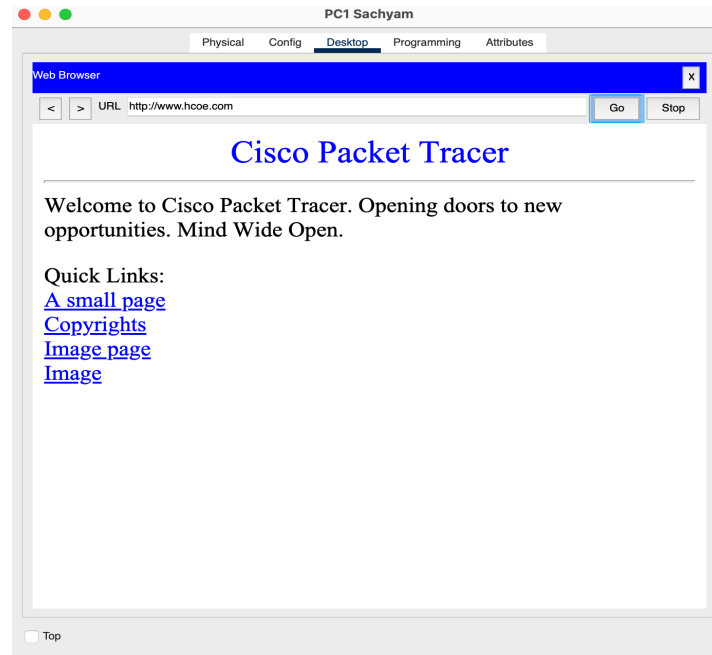


Fig: Tracing web server

DISCUSSION

In this experiment, a server was configured to deliver DHCP, DNS, and web services within a star-topology network using Cisco Packet Tracer. The DHCP service automatically assigned IP configurations to the client PCs, removing the need for manual setup. The DNS service successfully translated domain names into IP addresses, allowing clients to access the hosted web page using a browser. Overall, the integration of these services illustrated how centralized server management simplifies network administration and ensures smooth communication among devices.

CONCLUSION

This lab demonstrated the successful implementation of DHCP, DNS, and web services in a simulated network environment. DHCP streamlined IP address allocation, DNS enabled effective name resolution, and the web server provided accessible content to client devices. Together, these services highlighted the importance of centralized server configuration in creating efficient, manageable, and user-friendly network systems.